

HABIB UNIVERSITY

Engineering Mathematics MATH202 (3 Credit Hours)

Fall Semester - 2023

Instructor	Dr. Asgher Ali		
Class Timing and Location	Section L5:	Tu/Thu 05:20 – 06:35	FF-N220-SCB Classroom
Office Hours	Thursday – 4:00 PM to 5:00 PM		
Office Location	W-308		
Contact	asgher.ali@sse.habib.edu.pk		

Instructor	Dr. Salman Safdar		
Class Timing and Location	Section L3:	01:15 pm - 02:30 pm	E-012
	Section L6:	01:00 pm - 02:15 pm	E-012
Office Hours	Tuesday/Thursday 03:00 pm - 04:00 pm		
Office Location	TBA		
Contact	salman.safdar@sse.habib.edu.pk		

Instructor	Rameez Ragheb		
Class Timing and Location	Section L1:	MoWe 2:30PM - 3:45PM	W242
	Section L2:	TuTh 11:30AM - 12:45PM	E011
	Section L4:	Mo 11:30AM - 12:45PM	E011 – E012
Office Hours	Tu 3:30PM – 4:30PM, Fri 12:00PM-1:00PM		
Office Location	W-304		
Contact	rameez.ragheb@sse.habib.edu.pk		

Course LMS URL	https://hulms.instructure.com/courses/3941
Course Prerequisites	MATH 101, MATH 102
Content Area	This course meets math requirements for CS and EE programs as well as minor in Mathematics and Physics
Campus Safety Policy	Please read the campus safety policy and protocols if the classes are in-person

I. Rationale and Objective:

This course is designed to equip engineering and computer science students with some necessary mathematical tools and analytical reasoning methods that are an integral part of their future professions. Topics in the course have a wide range of applications such as, optimization problems, electrical circuit analysis, signal processing, mathematical modelling etc.

II. Student Learning Objectives:

i. CLOs

By the end of this course, students should be able to

CLO1. Determine vector field behavior by plotting and extend length of curve calculations to finding Line Integrals over parametrized curves in 3D vector fields.

CLO2. Calculate and estimate curl and divergence of vector fields

CLO3. Apply theorems of vector calculus to determine circulation and flux of a vector field.

CLO4. Solve first order ODEs by separability, exactness and integrating factor.

CLO5. Solve second order linear ODEs with constant coefficients.

CLO6. Apply Power Series and Laplace Transform methods to solve non-linear ODEs.

ii. Mapping of Quiz/Assignment/Exam with Course Learning Outcomes (CLOs)

iii. CLOs mapped to Program Learning Outcomes:

	Program Learning Outcomes (PLOs) mapped to MATH 202					
	CLOs of the course are designed to cater following EE PLOs: PLO 1 : Engineering Knowledge PLO 2 : Problem Analysis CLOs of the course are designed to cater following CS PLOs: PLO 1: Analysis PLO 6: Self-learning					
PLOs	Distribution of CLO weightages for each PLO					
	CLO 1	CLO 2	CLO 3	CLO 4	CLO 5	CLO 6
EE PLO 1						

EE PLO 2						
CS PLO 1						
CS PLO 2						

III. Format and Procedures:

The topics in the course will be taught through in-person lectures and demonstrations. Additionally, the course will have in class assignments and quizzes.

- Students are expected to be attentive in class and produce their own lecture notes.
- Students are expected to work up to 6 hours per week outside of the class for this course.

IV. Course Readings:

We will follow the following textbooks for this course:

Vector Calculus

Thomas' Calculus, Early Transcendentals by George B. Thomas, Maurice D. Weir and Joel R. Hass (thirteenth edition)

Referred henceforth as TC

Differential Equations

Advanced Engineering Mathematics by E. Kreyszig (tenth edition)

Referred henceforth as AEM

However students are welcome to follow the content through a book of their choosing. A library of recommended resources is shared and regularly updated on Canvas.

V. Assessments and Grading Procedures:

Assessments:

Short hand	Assessment	Occurrences	Total Weightage
HW	Homeworks	5	15
SQ	In-class short quizzes	12 (n-2)	20
CP	Class Participation	-	5
Project	Project	1	5
MID1	Midterm Exam	1	15
MID2	Midterm Exam	1	15
FIN	Final Exam	1	25

Note: For assessments with multiple occurrences, each occurrence has equal weightage.

Assessment Details:

- HW are to be completed individually and submitted on paper. Students are encouraged to discuss and solve the problems together but at no point should a student share their work with others in class or ask another student for their homework.

- CP is based on your presence, attentiveness, participation and behavior in the class room.
- Projects are group assignment. Each group should comprise of 4 to 5 students from the same section. Students are expected to form their groups by week 10. Project topics will be distributed in week 10 and students should submit a report on their project by week 15. Note that a topic that is assigned to a group may not be covered yet in week 10 (and may not be covered at all in the course), the objective of the project is self-learning in a team.
- In class quizzes would be unannounced, they would usually cover the topic that has been covered in the past week or whatever is happening in the class at present.
- Final Exam would be comprehensive and would cover the entire syllabus.

Late Submission Policy:

Students are expected to submit all assignments and activities on time. If the assignment is a late submission due to circumstances beyond their control, students may inform their instructor so that deadlines may be reset accordingly.

- For assignments and project report, a 10% grade reduction will be applied for every delay of 24 hours.
- Quizzes, Midterm and Final Exam must be submitted within the designated time.

Rubrics of Grading: All written instruments will follow the following assessing criteria

- The criteria for marking all assessment tasks will focus on correct working and appropriate reasoning in solutions and not just on correctness of answers.
- Consequently, a wrong attempt producing an answer coinciding with the correct answer **will be discarded**.
- No mistake will be penalized twice and a logical output of a mistake will be graded. This means error progression will be taken into account unless the student is making conceptual mistakes all over the solution.
- In the course you will learn variety of methods, some generally applicable and some for specific problems, and as long as the question does not restrict you to use a certain method you can use any method.
- When solving any problem, please make sure that your solution is well-written, clear and shows your conceptual understanding.
- If you use any new symbol to help you solve, please make sure to define that symbol explicitly.

GRADING SCALE

GRADING SCALE		
LETTER GRADE	GPA POINTS	PERCENTAGE
A+	4.00	[95-100]
A	4.00	[90-95)
A-	3.67	[85-90)
B+	3.33	[80-85)
B	3.00	[75-80)
B-	2.67	[70-75)

C+	2.33	[67-70)
C	2.00	[63-67)
C-	1.67	[60-63)
F	0.00	[0-60)

VI. Attendance Policy:

Students are expected to attend all sessions. Atleast 85% attendance in lectures is required to sit for the final exam. Students failing to meet the required criteria will be withdrawn from the course.

VII. Accommodations for students with disabilities

In compliance with the Habib University policy and equal access laws, I am available to discuss appropriate academic accommodations that may be required for student with disabilities. Requests for academic accommodations are to be made during the first two weeks of the semester, except for unusual circumstances, so arrangements can be made. Students are encouraged to register with the Office of Academic Performance to verify their eligibility for appropriate accommodations.

VIII. Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. Habib University is committed to providing an atmosphere for learning that respects diversity. While working together to build this community we ask all members to:

- share their unique experiences, values and beliefs
- be open to the views of others
- honor the uniqueness of their colleagues
- appreciate the opportunity that we have to learn from each other in this community
- value each other's opinions and communicate in a respectful manner
- keep confidential discussions that the community has of a personal (or professional) nature
- use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Habib community

IX. Office hours:

Office hours have been scheduled and circulated. During these hours the course instructor will be available to answer questions or provide additional help.

Every student enrolled in this course must meet individually with the course instructor at least twice during the semester. The first meeting should happen within the first four weeks of the semester but must occur before midterms. Any student who does not meet with the instructor may face a grade reduction or other penalties at the discretion of the instructor and will have an academic hold placed by the Registrar's Office.

X. Academic Integrity

Each student in this course is expected to abide by the Habib University Student Honor Code of Academic Integrity. Any work submitted by a student in this course for academic credit will be the student's own work.

Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by Habib University regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

PLAGIARISM: Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. As per University policy, plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated, bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.

CHEATING: The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.

COLLUSION: Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so. All violations of academic integrity will also be immediately reported to the Student Conduct Office.

You are encouraged to study together and to discuss information and concepts covered in lecture and the sections with other students. You can give "consulting" help to or receive "consulting" help from such students. However, this permissible cooperation should never involve one student having possession of a copy of all or part of work done by someone else, in the form of an e-mail, an e-mail attachment file, a diskette, or a hard copy.

Should copying occur, the student who copied work from another student and the student who gave material to be copied will both be in violation of the Student Code of Conduct.

During examinations, you must do your own work. Talking or discussion is not permitted during the examinations, nor may you compare papers, copy from others, or collaborate in any way. Any collaborative behavior during the examinations will result in failure of the exam, and may lead to failure of the course and University disciplinary action.

Penalty for violation of this Code can also be extended to include failure of the course and University disciplinary action.

XI. Tentative week wise schedule

Week	Topic(s)	Specific topics for the week and Relevant topics and sections from the course book.	Remarks
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Week 1	<u>Theorems of Vector Calculus and their applications.</u> Vector fields, Line Integrals	• Vector Fields • Line Integral, Work, Circulation TC 16.2	
Week 2	<u>Theorems of Vector Calculus and their applications.</u> Fundamental Theorem of Vector Calculus	• Path Independence, Conservative Vector Fields • Gradient Fields • Potential Function • Thm: Conservative iff Gradient Field TC 16.3	HW1-released at the end of week
Week 3	<u>Theorems of Vector Calculus and their applications.</u> Curl, Divergence	• Physical and mathematical concepts of Curl and Divergence of vector field. • Calculating the Curl and Divergence of Vector fields (in 2D and 3D Cartesian coordinates) TC 16.4	HW1- due at end of week
Week 4	<u>Theorems of Vector Calculus and their applications.</u> Greens' Theorem and its applications.	• Derivation of Greens Theorem • Application of Greens theorem in simplifying problems. • Introduction to Surfaces • Parametrization of Surfaces and Normal Vectors TC 16.4, 16.5	HW2- released of week
Week 5	<u>Theorems of Vector Calculus and their applications</u> Divergence theorem and its application.	• Flux Integral • Derivation of the Divergence theorem. • Physical meaning of the Divergence theorem in terms of flux of the field. TC 16.6, 16.8	
Week 6	<u>Theorems of Vector Calculus and their applications</u> Stoke's theorem and its application.	• Derivation of the Stoke's theorem. • Application of Stoke's Theorem in simplifying problems. TC 16.7	HW2- due HW3 – released

Week 7	<u>Ordinary Differential Equations (ODEs)</u> Introduction to Ordinary Differential Equations (ODEs)	- Categories of differential equations (linear-nonlinear, homogeneous-inhomogeneous and order) - Introduction to Ordinary Differential equations (ODEs). - Initial Value Problems (IVPs) in first order ODEs. AEM 1.1, 1.2	HW3- due
			Midterm 1 (All topics covered till Week 6)
Week 8	<u>Ordinary Differential Equations (ODEs)</u> First Order ODEs.	- Solving first order separable ODEs (and ODEs that can be converted into separable ODEs). - Exact and In-Exact ODEs and their solutions. AEM 1.3, 1.4	
Week 9	<u>Ordinary Differential Equations (ODEs)</u> Solution of Exact and Inexact First Order ODEs	- Inexact ODEs which can be converted to exact ODEs. - Solution of general first order linear ODEs. - Existence and Uniqueness of solution of first order ODE. AEM 1.5	
Week 10	<u>Ordinary Differential Equations (ODEs)</u> Existence and uniqueness of solution of first order ODE.	- Introduction to second order ODEs. - Categories of Second order ODEs (linear-nonlinear, homogeneous-inhomogeneous). - Nature of the solution (Superposition of solutions) AEM 2.1, 2.2	
Week 11	<u>Ordinary Differential Equations (ODEs)</u> Second order ODEs	- Solving linear second order ODEs with constant coefficients. - Initial Value Problems in second order ODEs. - Solution of Inhomogeneous second order ODEs using Undetermined Coefficient Methods. - Existence and Uniqueness of solution of second order ODE. AEM 2.6, 2.7	HW4-released Nov 4th
Week 12	<u>Ordinary Differential Equations (ODEs)</u> <u>Solution of ODE</u>	- Basics of what a mathematical series - Introducing Power series and, its application, Taylor series. - Convergence of a series and radius	Midterm 2

	<u>using Power Series</u> Theory of using the Power series to solve ODEs.	- of convergence. - Concepts of analytic functions, regular points and singular points. AEM 5.1	
Week 13	<u>Ordinary Differential Equations (ODEs)</u> <u>Solution of ODE using Power Series</u> Method of solving an ODE using Power series	- Using Power series to solve an ODE around a regular point. - Problems in series solution of ODE. AEM 5.2	HW4- due HW5-released
Week 14	<u>Ordinary Differential Equations (ODEs)</u> Solution of ODE using Laplace Transform	- Introduction of Laplace Transforms - Taking Laplace transform of ODEs. AEM 6.1, 6.2	HW5-due
Week 15	<u>Ordinary Differential Equations (ODEs)</u> Solution of ODE using Laplace Transform	- Solving ODEs using Laplace transform. - Solving ODEs with Impulse function and Unit step function - Review of ODE solutions using different methods. AEM 6.3, 6.4	
	Reading Days		
	End Term Examinations Days		
	Faculty to discuss semester grades with students		
	Final Grades Due by the Faculty: Fall 2024		

Note: University's Examination Policy available in the Academic Policies folder on the Faculty / Staff Portal.